HIMCM TRYOUT PROBLEM & GUIDELINES

MMC Officers

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1 Guidelines

If you already have an idea of how to write your paper, feel free to skip past this section.

1.1 Formatting

1. **Define the problem** (who is the audience, who is affected, what data matters?). What do variables in the problem even mean? How could you define variables that relate to the problem? For example, a past problem asked to predict total metals used by computers in the next 50 years. Should you predict by mass, by volume, and which metals are the most important to consider?

2. **State assumptions** (and justify). What assumptions are you making that make the problem actually solvable or meaningful? For example, population isn't constant, but it is approximately constant over a year, which can help. The Earth isn't a sphere, but it's approximately one.

3. **Describe the model** Explain what model you used, and **WHY**. A good model is the model with the best justification, not the most sophisticated one. Don't use a chainsaw instead of a screwdriver because it looks cooler.

4. **Analyze** What results did you have? Why do you think you had them? What did you do right, what didn't you do right? ($\tilde{4}.5$) **do a sensitivity analysis** How well does your model pertain to real world data or just overfit? Is it reasonable for more than just a narrow window that you use? For example, instead of predicting 30 years, try 5, or 50. This is like that extra sophistication point.

5. **Strengths and Weaknesses of Model** This ties in to both sections 4 and 3, but this is more of a retroactive analysis. You are seeing how the parameters and the way the model functioned made it stronger or weaker. Ask what you would improve. What trade-offs did you make?

6. **Appendix** THIS DOES NOT COUNT TO THE 3 PAGE LIMIT. Include a compilation of code, data sets, PROMPTS (if using an LLM), and other info. You don't need citations, but your sources should be included.

1.2 Additional stuff

Prose Suggestions: While you don't HAVE to write in perfect English, it is ABSOULUTELY viewed fondly to write in an eloquent manner. Don't write verbosely, either.

Graphing Stuff: If you can include a pretty matplotlib or other fun graph, please do. Judges love seeing a nice visual.

Tables: Use tables. Tables are aesthetic and a nice way to show information about variables and data.

Example Paper: 2023 TJMMC Paper

2 Problems

Pick one of these two problems. Put the problem number you chose at the top of your paper, so that we know which you picked.

2.1 Data Centers in Loudon County



Figure 1: Problem 1

While not a native Loudon resident, I was recently driving by Loudon County and was astounded by the sheer volume of data centers. I don't usually think about the environment, but I recently heard that Lake Fairfax was continually heating due to data centers, and I like cold water. So I wonder what the impact of the Loudon County data center expansion will be. Help me answer this by

addressing the following:

What will be the total energy consumption over the next 10 years of the data centers in Loudon County?

There has been a rise in power consumption due to AI and computational systems' rapid expansion. See if you can attribute the rise in technology to power consumption and data center expansion. Here are some sources to help your research.

- JLARC: Data Centers in Virginia (2024)
- Lawrence Berkeley National Lab: 2024 U.S. Data Center Energy Usage Report
- International Energy Agency: Energy and AI (2025)
- IEA: Data Centre Electricity Use Surged in 2025
- Loudoun County: Data Centers, Energy Issues & Considerations
- Loudoun County Data Center Brief (Supervisor Turner, 2026)
- Piedmont Environmental Council: Data Centers in Loudoun, A Primer
- Broadband Breakfast: Data Centers Drive New Energy Disputes in Northern Virginia
- Northern Virginia Magazine: Data Centers Could Cause Heat Islands and Water Shortages
- FFXnow: Fairfax Environmental Council Worried About Data Centers' Water Use

2.2 Damage caused by spotted lantern flies



Figure 2: Problem 2

I don't know if you've been outside recently, but lately spotted lanternflies are everywhere. They've vastly repopulated as an invasive species over a short period of time, and I want to measure just how much harm they will cause if this trend keeps up. To help me with this, answer the following:
What will be the predicted damage to the tree ecosystem in Northern Virginia due to spotted lantern flies over the next 20 years?
Examine their expansion, to what extent they harm the ecosystem, and other factors when writing your paper. Consider how you will measure tree damage. Here are some sources to start your research.

- Urban & Leach: Biology and Management of the Spotted Lanternfly (Annual Review of Entomology, 2023)
- Lavely et al.: Impacts of Spotted Lanternfly Feeding on Young Hardwood Trees (2022)
- Penn State: Feeding Damage Caused by Spotted Lanternflies on Young Maples
- Ladin et al.: Human-Mediated Dispersal Drives the Spread of the Spotted Lanternfly (2023)
- Madalinska et al.: Locating Spotted Lanternfly Egg Masses in Wooded Habitat (2022)
- Philadelphia Inquirer: How Far Can Spotted Lanternflies Spread in a Year?

- Utah State University Extension: Spotted Lanternfly Biology
- Northern Virginia Magazine: Spotted Lanternflies in NoVA
- Fairfax County: Spotted Lanternfly Seen in More Areas
- Fairfax County: Tree of Heaven Brochure